

VIRIKA OLIVIA SOANS

virika06@gmail.com • +91 99863 02506 • [linkedin.com/in/virika-olivia-soans-3727122b8](https://www.linkedin.com/in/virika-olivia-soans-3727122b8) • github.com/V-virika • Bengaluru, India

PROFESSIONAL SUMMARY

AI/ML-focused technology enthusiast with expertise in Python, SQL, data analytics, and software development. Skilled in designing and implementing efficient solutions, analyzing complex datasets, and developing applications that drive informed decision-making. Experienced in data processing, database management, and software engineering principles, with a strong foundation in problem-solving, algorithmic thinking, and delivering quality technical solutions. Well-versed in artificial intelligence and machine learning, with hands-on experience in building predictive models, applying statistical techniques, and leveraging ML algorithms to uncover insights and optimize performance.

CORE COMPETENCIES

Analytics & ML: Predictive Modelling, Statistical Analysis, Data Visualization, Deep Learning, Explainable AI

Tools & Platforms: Python, SQL (MySQL), Power BI, Jupyter, GitHub, Cloud Platforms

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, Gradio, Flask

Other: HTML, CSS, Frontend Development

PROJECTS & RESEARCH

Deep Learning-Based Prostate Cancer Classification

Vision Transformers, Explainable AI

- Engineered a Swin Transformer pipeline with CLAHE preprocessing, achieving 96.4% classification accuracy and 0.90 ROC-AUC on histopathological images — outperforming CNN baselines by ~8%.
- Integrated Gradio-based explainable AI visualization to highlight diagnostic decision regions, enabling clinician trust and supporting scalable hospital deployment.

Finova AI – Financial Stability & Scheme Recommendation Platform

React, Node.js, PostgreSQL, Firebase, AI-based Risk Analysis

- Developed an AI-driven platform to detect early debt risk and map users to relevant government schemes (PMJDY, Mudra, SHG, MSME), enabling financial inclusion for underserved communities.
- Built a responsive React frontend with seamless API integration to a Node.js backend, enabling real-time financial analysis, risk scoring, and personalized guidance.

IMPRINTS — Fingerprint-Based Blood Group Identification

EfficientNet, Web App

- Built a non-invasive blood group prediction system using EfficientNet, achieving 85.84% accuracy — deployed as a real-time web app for emergency and rural healthcare settings.
- Designed for scalability in resource-constrained environments, reducing reliance on lab tests in underserved communities.

Smart Agriculture IoT System

NodeMCU, IoT, Real-Time Analytics | 1st Place, University Expo 2024

- Deployed IoT sensor network + automated irrigation system with real-time soil moisture analytics via Blynk, reducing manual oversight and optimizing water usage.
- Won 1st place at Dayananda Sagar University project expo (2024)

Hybrid Piezo-Solar Smart Road for Sustainable EV Charging

ESP32, IoT, Wireless Power Transfer, Renewable Energy | Academic Project

- Deployed IoT sensor network + automated irrigation system with real-time soil moisture analytics via Blynk, reducing manual oversight and optimizing water usage.
- Won 1st place at Dayananda Sagar University project expo (2026)

AEGIS.AI — Secure AI Prompt Firewall & Data Protection Platform *Prompt Firewall, DLP & Sensitive Data Detection, Prompt Injection, Risk Scoring, Data Masking, RBAC*

- Developed a real-time AI prompt firewall to detect sensitive data, malicious prompts, and policy violations before requests reach LLMs. Built an integrated security platform with RBAC, STEGOShield secure communication, risk analysis, data masking, and centralized audit monitoring

Adaptive Audio Steganography (STC + GOAS)

2nd Place, University Expo 2025

- Designed a high-fidelity steganography system using Syndrome-Trellis Codes and Hamming coding, embedding encrypted messages in audio with robust imperceptibility metrics.

PUBLICATIONS

- "Multimodal Machine Learning Approaches for Early Detection and Classification of Breast Cancer Using Imaging and Genomic Data" — Published 2025
- "Automated License Plate Recognition Using Deep Learning for Smart Mobility" — Published 2026
- "Hybrid Piezo-Solar pad with wheel embedded receiver for sustainable EV charging" — Published 2026

EDUCATION

Dayananda Sagar University

B.Tech — Computer Science & Engineering (Data Science) 2026 | **CGPA: 9.42 / 10**

Christ Academy Junior College

Class XII (PCMB) 2022 | **89.3%**

De Sales Academy

Class X (ICSE) 2020 | **92%**

ACHIEVEMENTS & AWARDS

- 1st Place — Project Expo, Smart Agriculture IoT System, Dayananda Sagar University (2024)
- Global Nominee — NASA Space Apps Challenge 2024 ('Leveraging Earth Observation Data for Informed Agricultural Decision-Making')
- 2nd Place — Project Expo, Adaptive Audio Steganography, Dayananda Sagar University (2025)
- Best SDG Innovation — 36 Hours Hackathon-CodeSangram, Alliance University (2026)